

Legal Assistant System

Complete Architecture & Workflow Analysis

System Overview

Case Discovery

Charge Analyzer

Language Assistant

Summary Generator

Citation Validator

Petition Draft

Core Features

Intelligent Case Discovery

Semantic search across legal precedents

Charge Pattern Analyzer

Predicts conviction probability & patterns

Language Intelligence Assistant

Improves legal writing & citations

Case Summary Generator

Auto-generates judgment summaries

Citation Validator

Verifies and validates legal references

Petition Drafting Assistant

Auto-drafts legal petitions

System Architecture

Frontend Layer

React Dashboard

Tailwind CSS

File Upload

PDF Viewer

API & Processing Layer

FastAPI/Flask

PDF Parser

NLP Pipeline

Authentication

AI/ML Layer

OpenAI GPT-4

Sentence Transformers

Custom Fine-tuned Models

Embedding Models

Database Layer

PostgreSQL

Vector DB (Pinecone)

Redis Cache

File Storage

Overall System Data Flow

Input Sources

PDFs, Court DBs, User Queries

→

PDF Processing

Parse, Extract, Normalize

→

Data Storage

Structured + Vector DBs

→

AI Processing

LLMs + ML Models

→

User Interface

Dashboard + Results

Legal Assistant System

Complete Architecture & Workflow Analysis

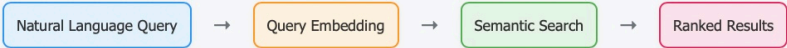
System Overview **Case Discovery** Charge Analyzer Language Assistant Summary Generator Citation Validator Petition Draft

Case Discovery Engine Workflow

Training Phase:



Usage Phase:



Databases

Pinecone (Vector DB)
PostgreSQL Redis Cache

AI/ML

Sentence-BERT
Custom Embeddings
Similarity Search

Processing

PyPDF2 spaCy NLP
Regex Patterns

APIs

FastAPI RESTful API
WebSocket

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Case Summary Generator Workflow

Judgment PDF



Extract Structure



GPT-4 Summarization



Timeline Extraction



Structured Summary

Storage

File Storage

Summary Cache

PostgreSQL

AI/ML

GPT-4 API

Summarization Models

NER Models

Processing

PDF Text Extraction

Structure Recognition

Date/Timeline Parsing

Output

2-Para Summary

Key Points

Timeline

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Citation Validator Workflow

Draft with Citations



Extract Citations



Database Lookup



Validity Check



Validation Report

Databases

Citation Database

Overruled Cases DB

PostgreSQL

Processing

Regex Patterns

NER Models

Text Parsing

Validation

Citation Format Check

Currency Verification

Authority Check

Features

Error Detection

Alternative Suggestions

Validation Scoring

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Petition Drafting Assistant Workflow

Case Facts & Charges



Template Selection



GPT-4 Generation



Citation Integration



Draft Petition

Databases

Template Database

Phrase Bank

Citation DB

AI/ML

GPT-4 API

Template Matching

Legal Style Model

Processing

Fact Extraction

Charge Mapping

Document Assembly

Output

Structured Petition

Auto-Citations

Legal Formatting